universe is 14 billion years old, and if so, what would have ensured that their clocks have been ticking away in synch with ours? The answer relies on symmetry—symmetry in space.

If your eyes could see light whose wavelength is much longer than that of orange or red, you would not only be able to see the interior of your microwave oven burst into activity when you push the start button, but you would also see a faint and nearly uniform glow spread throughout what the rest of us perceive as a dark night sky. More than four decades ago, scientists discovered that the universe is suffused with microwave radiation—long-wavelength light—that is a cool relic of the sweltering conditions just after the big bang.[4] This cosmic microwave background *radiation* is perfectly harmless. Early on, it was stupendously hot, but as the universe evolved and expanded, the radiation steadily diluted and cooled. Today it is just about 2.7 degrees above absolute zero, and its greatest claim to mischief is its contribution of a small fraction of the snow you see on your television set when you disconnect the cable and turn to a station that isn't broadcasting.

But this faint static gives astronomers what tyrannosaurus bones give paleontologists: a window onto earlier epochs that is crucial to reconstructing what happened in the distant past. An essential property of the radiation, revealed by precision satellite measurements over the last decade, is that it is extremely uniform. The temperature of the radiation in one part of the sky differs from that in another part by less than a thousandth of a degree. On earth, such symmetry would make the Weather Channel of little interest. If it were 85 degrees in Jakarta, you would immediately know that it was between 84.999 degrees and 85.001 degrees in Adelaide, Shanghai, Cleveland, Anchorage, and everywhere else for that matter. On a cosmic scale, by contrast, the uniformity of the radiation's temperature is *fantastically* interesting, as it supplies two critical insights.

First, it provides observational evidence that in its earliest stages the universe was not populated by large, clumpy, high-entropy agglomerations of matter, such as black holes, since such a heterogeneous environment would have left a heterogeneous imprint on the radiation. Instead, the uniformity of the radiation's temperature attests to the young universe being homogeneous; and, as we saw in